Veterinary Parasitology 217 (2016) 7–13

Contents lists available at ScienceDirect

Veterinary Parasitology

journal homepage: www.elsevier.com/locate/vetpar

Research paper

Assessing the risk of bovine fasciolosis using linear regression analysis for the state of Rio Grande do Sul, Brazil

Ana Elisa Pereira Silva[a], Corina da Costa Freitas[b], Luciano Vieira Dutra[b], Marcelo Beltrão Molento[c,d,*]

[a] Centro Nacional de Monitoramento e Alertas de Desastres Naturais, CEMADEN, Estrada Doutor Altino Bondesan, 500, Distrito de Eugênio de Melo, CEP 12.247-016, São José dos Campos, SP, Brazil
[b] Instituto Nacional de Pesquisas Espaciais, INPE, Av. dos Astronautas, 1758, Jd. Granja, CEP 12.227-010, São José dos Campos, SP, Brazil
[c] Universidade Federal do Paraná, UFPR, Rua dos Funcionários, 1540, CEP: 80 035 050, Cabral Curitiba, PR, Brazil
[d] Instituto Nacional de Ciência e Tecnologia, INCT-Pecuária. Av. Antônio Carlos, 6627 Belo Horizonte, MG, Brazil

ARTICLE INFO

Article history:
Received 8 July 2015
Received in revised form
19 December 2015
Accepted 21 December 2015

Keywords:
Parasitic diseases
Cattle
Remote sensing technology
Geographic information systems
Regression analysis

ABSTRACT

Fasciola hepatica is the causative agent of fasciolosis, a disease that triggers a chronic inflammatory process in the liver affecting mainly ruminants and other animals including humans. In Brazil, *F. hepatica* occurs in larger numbers in the most Southern state of Rio Grande do Sul. The objective of this study was to estimate areas at risk using an eight-year (2002–2010) time series of climatic and environmental variables that best relate to the disease using a linear regression method to municipalities in the state of Rio Grande do Sul. The positivity index of the disease, which is the rate of infected animal per slaughtered animal, was divided into three risk classes: low, medium and high. The accuracy of the known sample classification on the confusion matrix for the low, medium and high rates produced by the estimated model presented values between 39 and 88% depending of the year. The regression analysis showed the importance of the time-based data for the construction of the model, considering the two variables of the previous year of the event (positivity index and maximum temperature). The generated data is important for epidemiological and parasite control studies mainly because *F. hepatica* is an infection that can last from months to years.

© 2015 Elsevier B.V. All rights reserved.

1. Introduction

Fasciolosis is a disease caused by *Fasciola* sp. parasites (*Fasciola hepatica* and *F. gigantica*) whose host involves an array of animals, including humans. In Europe, Americas and Oceania, only *F. hepatica* is found, whereas in Africa and Asia both species are present. No continent is free from fasciolosis and as suggested in a report by the WHO (2014) it is likely that the identification of infected animals also point out to infected humans. Nonetheless, infected humans may maintain the infection in a certain area by releasing eggs in the environment (pasture, agriculture) shared by animals, configuring its anthropozoonotic status.

The life cycle of *F. hepatica* and the transmission of the disease depend on environment conditions (Oliveira and Filha, 2009). In Brazil, *Lymnaea columella* and *L. viatrix* are the intermediate hosts for *F. hepatica*, where animals get the infection by ingesting the metacercariae while pasturing (Toledo et al., 2012). The transmission to humans occurs in a similar condition of infection with the ingestion of contaminated vegetables. Even though the prevalence of animal fasciolosis may facilitate the transmission and dissemination of the disease among humans, Mas-Coma et al. (1999) reported that a high prevalence of fascioliasis in humans does not necessarily occur in areas where *F. hepatica* is a major veterinary problem.

Although most of the cases described in livestock are still being identified in the South of the country (Dutra et al., 2010) fasciolosis is presently expanding throughout Brazil (Aleixo et al., 2015). Geographic information system, GIS has been used as an exploratory spatial analysis to assess the distribution of *F. hepatica* (Bennema et al., 2014; Freitas et al., 2014; Aleixo et al., 2015), including climatic and topographical data where inear regression is a statistical method that analyzes the relation between two or more variables (quantitative or qualitative) in which the dependent variable is explained by one or more independent variables (Neter et al., 1996). The study of climatic and environmental data associated with different statistical analysis to interpret *F. hepatica* risk areas (Martins,

* Corresponding author. Fax: +55 4133505623.
E-mail addresses: anaepsilva@hotmail.com (A.E.P. Silva), corina@dpi.inpe.br (C.d.C. Freitas), dutra@dpi.inpe.br (L.V. Dutra), molento@ufpr.br (M.B. Molento).

http://dx.doi.org/10.1016/j.vetpar.2015.12.021
0304-4017/© 2015 Elsevier B.V. All rights reserved.

2008; Fonseca, 2009; Dutra et al., 2010; Valencia-López et al., 2012; Ducheyne et al., 2015; Selemetas and Waal, 2015) is essential to visualize the dynamics of the disease and the external factors that could influence its occurrence. Ducheyne et al. (2015) using a harmonized sampling approach in combination with spatial modeling (rainfall and temperature) assessed the distribution of fasciolosis in dairy cattle in Europe (Belgium, Germany, Ireland, Poland and Sweden) in association with the data from an indirect bulk tank milk (BTM) enzyme-linked immune sorbent assay (ELISA). The authors identified rainfall and temperature as the most important factors for probability of exposure when 571 municipalities were categorized as positive and 429 as negative. This paper aims to establish the relationship between the positivity index (Pi) of bovine fasciolosis and the climate and environment variables in an endemic area in order to use the data to assess the risk of the disease occurrence using multiple linear regression methods.

2. Material and methods

2.1. Study area and data collection

The study area (Fig. 1) encompasses 497 municipalities of Rio Grande do Sul, South of Brazil. The cities were selected with data from cattle producers that sent their animals to slaughterhouses inspected by the Brazilian Federal Meat Inspection Service of the Ministry of Agriculture, MAPA/SIF 2354.

The information on the number of slaughtered, and negative or infected animals with fasciolosis was collected monthly and per municipality. The rate of infected animal per slaughtered animal for each city, hereby denominated positivity index (Pi) for the disease, was calculated through the equation:

$$\mathrm{Pi} = \frac{\mathrm{Ni}}{\mathrm{Ns}} \times 100 \quad (1)$$

where Ni and Ns are respectively the number of infected and slaughtered animals for each city. The Pi for each city was used as a dependent variable for the construction of the regression model. The inspected municipalities which did not have fasciolosis cases at the inspection were also included as part of the sample. With the data on the fasciolosis Pi distribution in each year, municipalities were classified as low (<8%), medium (8–18%) and high (>18%) risk groups. The analysis did not permit 0% of infection rate.

2.2. Environmental and climatic variables

Climate and environment variables related to the development of both *F. hepatica* and the *Lymnaea* sp. snail were used as independent variables for the construction of the regression model. The Pi for earlier years (Pi_ey) was also included as an independent variable to improve the generation of the model. Independently from climate and environmental conditions, fasciolosis will not manifest unless there is the presence of its etiologic agent and its intermediate host. Table 1 display the climate and environment variables employed in the construction of the model.

Altitude (Alt_ave) was extracted from a continuous vectorial cartographic base, elaborated by the Federal University of Rio Grande do Sul, UFRGS (Hasenack and Weber, 2010). Through the sampled points in each municipality, a topological operation was made, resulting in the average altitude for all the municipalities in the state. The percentage of water-covered areas (Water_perc) in each municipality was obtained from the same cartographic database. The hydrographic vector file was changed to a matrix format and then the water accumulation was obtained for each municipality. The values for accumulation potential and the total city area allowed the calculation of the percentage of water accumulation potential for each municipality.

The slope variable was acquired through remote sensoring images from the SRTM (Shuttle Radar Topography Mission), providing the elevation data in a global scale. Through the use of *Topodata*, a Brazilian Geomorphometric Database (http://www.dsr.inpe.br/topodata), the slope numerical product for the entire state was obtained followed by the construction of the average slope for each municipality.

The EVI (enhanced vegetation index) and NDVI (normalized difference vegetation index) variables, precipitation and temperature, which showed the seasonal variation between summer and winter had their data interval selected from the vegetation indexes obtained by the MODIS sensor (http://modis.gsfc.nasa.gov/). In order to extract such data, two MODIS images were used (H13V11 and H13V12). Based on a preliminary study of the EVI and NDVI images from 2006, two acquisitions periods have been chosen for representing winter and summer data. The year of 2006 was chosen due to the non-incidence of the climate event known as *La Niña* and the minimal effects of *El Niño*. Such events may change significantly the air temperature and the rainfall regimen in Southern South America, including Brazil. The dates selected were January 17 and February 01 for the summer and August 29 and September 13 during winter, periods in which the index values for vegetation were the highest in the summer and the lowest in the winter of that specific year. Each 15-day period corresponded to an image from which the vegetation index value was extracted.

Precipitation and air temperature were acquired from the Weather Forecast and Climate Study Center at the National Institute for Space Research, CPTEC/INPE through data from the Data Collection Platforms (PCDs) interconnected in a 0.25° grid point for Brazil. The extracted values were the ones closest to the geographic localization of the municipalities studied.

Precipitation was accumulated for the same period of the dates chosen for the MODIS images, from which the vegetation index values were extracted. Like the precipitation data, the minimum and maximum temperature information were obtained for the same 15-day period and the average for each variable and period was extracted.

Even though our study used data from the State and municipal level and not by farms or regions we understand the potential influence of different types of soil and pasture management for the same municipality. In this regard, we assume that the vegetation index would account for the estimation and be a valuable factor.

2.3. Linear regression analysis

A multiple linear regression model was built using the dependent variable (Pi), submitted to a logarithmic transformation (LnPi) in order to increase its correlation to the independent variables. On average, about 300 samples were used per year of the study from 2003 to 2010. The samples were randomly divided in: 1. for the model construction (2/3 of the samples) and 2. for the model validation (1/3 of the samples).

By using a correlation matrix, the variables that were weakly correlated to the disease's Pi or not significant were removed. The removed variables were EVI_win, NDVI_sum, NDVI_win, Prec_win, Tmax_sum and Tmin_win. All the other variables were included in the model along with the data for all previous years.

Individual models for each year were generated and the one with the highest coefficient of determination (R^2) and the least variables were selected. Because the model chosen for each year presented almost the same independent variables, we built a single model generated from these variables, using samples from all years. Therefore, a general model was developed concatenating the data of all years.

The general model was determined with 1.178 samples selected for model construction and 595 samples for model validation. The

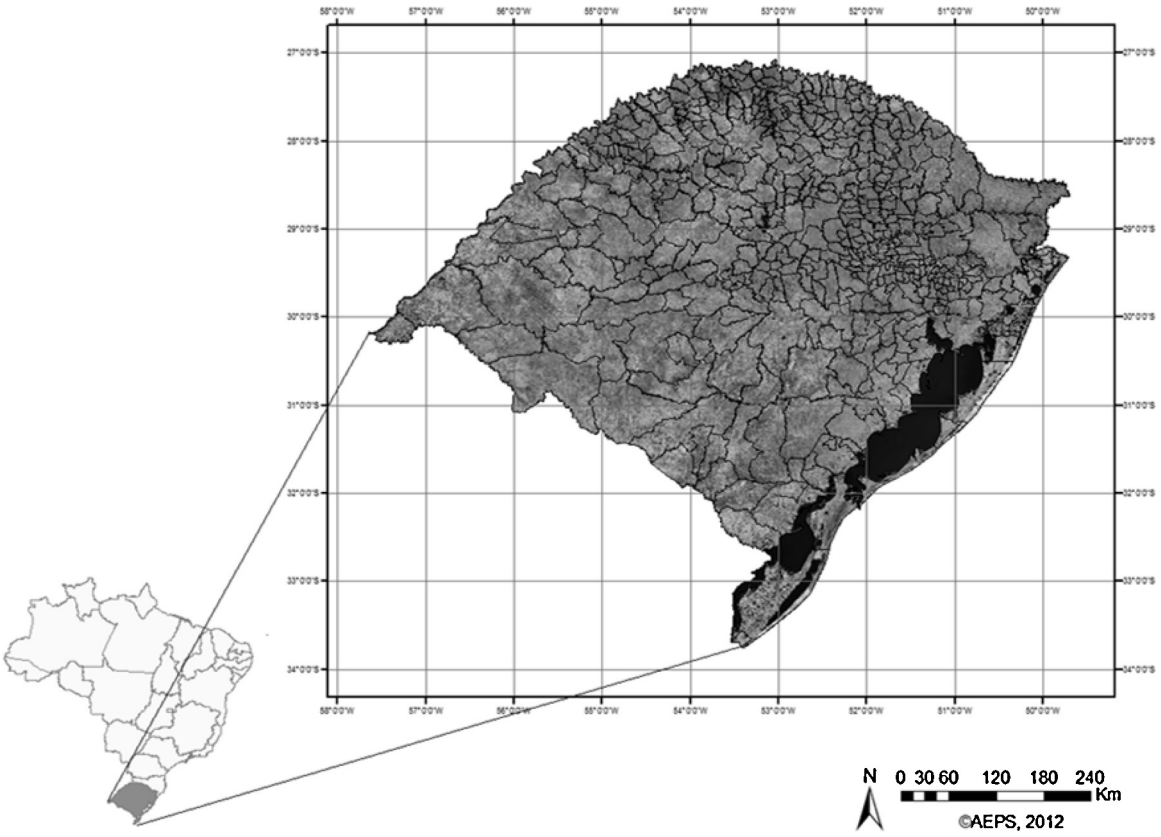


Fig. 1. Location of the State of Rio Grande do Sul, Brazil, divided by its city limits. The two large black areas are lakes, close to the Atlantic Ocean.

model generated through the multiple linear regression analysis is described in the equation below:

$$\mathrm{Pi}_{\mathrm{general}} = e^{(2.736+0.401\times\mathrm{LnPi_ey}-0.001\times\mathrm{Alt_ave}-0.044\times T_{\mathrm{max}}\mathrm{_win_ey})} - 1 \quad (2)$$

where e is exponential, LnPi_ey is the logarithmic transformation of the positivity index for the previous year, Alt_ave is the average altitude and T_{max}_win_ey is the average maximum temperature in the winter for the previous year.

3. Results

The years 2007 and 2008 were affected by the climate event *La Niña*, whose intensity was regarded as significantly high for those years. The event declined the average temperatures and lowered the overall precipitation volumes. Therefore, we believe that such event may have affected the generated models for the years that used the variables from 2007 and 2008 as input data. So, the general model was created based on samples from all years, except 2008 and 2009, as their input variables came from the *La Niña* years.

As expected, Pi of one year was directly proportional for the Pi of earlier years. Nevertheless, the variables Alt_ave and T_{max}_win_ey, which was the maximum temperature from the winter of earlier years, presented an inverse correlation ($R = -0.53$ and -0.45, respectively) to the disease, i.e., the lower their value, the greater the Pi for *F. hepatica* occurrence (data not shown).

The general model presented a coefficient determination of 0.36. The general model also met the validation requirements (Neter et al., 1996) as the values for the residual mean square (RMS = 0.77) and the mean square prediction error (MSPR = 0.81) were close and had the same order of magnitude. Therefore, the constructed model

Table 1
Environmental and climatic variables used to generate the prediction model for *F. hepatica* in Rio Grande do Sul state, Brazil.

Variable	Acronym	Variables description	Source[a]
Environmental	Alt_ave	Average altitude	UFRGS
Environmental	Water_perc	Percentage of water accumulation potential	UFRGS
Environmental	Slope_ave	Average slope	SRTM
Environmental	EVI_sum	Enhanced vegetation index in the summer[b]	MODIS
Environmental	EVI_win	Enhanced vegetation index in the winter[b]	MODIS
Environmental	NDVI_sum	Normalized difference vegetation index in the summer[b]	MODIS
Environmental	NDVI_win	Normalized difference vegetation index in the winter[b]	MODIS
Climatic	Tmin_sum	Average minimum temperature in the summer[b]	CPTEC
Climatic	Tmin_win	Average minimum temperature in the winter[b]	CPTEC
Climatic	Tmax_sum	Average maximum temperature in the summer[b]	CPTEC
Climatic	Tmax_win	Average maximum temperature in the winter[b]	CPTEC
Climatic	Prec_sum	Accumulated precipitation in the summer[b]	CPTEC
Climatic	Prec_win	Accumulated precipitation in the winter[b]	CPTEC

[a] UFRGS—Federal University of Rio Grande do Sul, SRTM—Shuttle Radar Topography Mission, MODIS—Moderate Resolution Imaging Spectroradiometer, CPTEC—Weather Forecast and Climate Studies Center.
[b] Between years 2002 and 2010.

Table 2
Beta coefficients and *p*-values of the variables in the general model for the prediction of the occurrence of *Fasciola hepatica* in Rio Grande do Sul, Brazil.

Variables	Beta coefficients	*p*-values
LnPi_ey	0.415	0.001
Alt_ave	−0.263	0.001
T_{max}_win_ey	−0.098	0.001

was statistically adequate to estimate the risk for bovine fasciolosis in the state of Rio Grande do Sul.

Due to a difference in the measurement units between the model variables, the Beta coefficients were used to compare to the regression coefficients (Neter et al., 1996). Thus, the higher the Beta coefficient values in the model, the most relevant were the variables. The Beta coefficient and the *p*-value for the variables in the general model are presented in Table 2, ranked by relevance.

Using the general model (Eq. (2)) it was possible to assign a label of infection severity to each municipality (Section 2.1) and then compare the predicted severity degree with the real ones from known places. This comparison may be described as a contingency matrix, known as a confusion matrix. A set of confusion matrixes was obtained for each year (Table 3). The user accuracy showed the percentage of samples correctly classified in each class, ranging from 48 to 66%. Although *F. hepatica* is not a common parasite in most of the country with averages below 1.5%, most of the state of Rio Grande do Sul can be regarded as a high endemic area. The overall accuracy was calculated as the ratio of corrected classified samples over the total amount of samples for every year (Table 3). The non-pathogenic samples were also classified for all years and most of them were correctly classified as low risk cities (Table 4). The presented data showed that 76% of the samples with low residuum, ranging between -8 and 8%, where the most exact model found for the samples. There was a total of 5% of overestimated samples and 19% of underestimated samples.

All municipalities have been classified as having low, medium or high Pi. Figs. 2 and 3 show two of eight of the thematic maps, where Pi is represented as the observed fasciolosis data (Figs. 2a and 3a) or the estimated values (Figs. 2b and 3b) from the general model. The data on Fig. 2 is for 2005 analysis and represent the least correctness (48%), as 2006 is the most accurate classification (66%) (Fig. 3). Figs. 2a and 3a also show the non-pathogenic samples, most in the North of RS (chess-board areas), which are the ones that did not report any *F. hepatica*. The maps for the years 2003, 2004, 2007, 2008, 2009 and 2010 can be accessed as additional files or be requested with the corresponding author.

4. Discussion

The most important variable for the general model was the LnPi_ey, followed by Alt_ave and T_{max}_win_ey, showing that if a certain municipality had already sent bovines for slaughtering and the presence of the *F. hepatica* was detected in these animals, the chance for new local occurrence of fasciolosis in the following year was higher than the ones that had never presented the infection. This fact is epidemiologically possible by the presence of positive animals and the over season pasture contamination. Knubben-Schweizer et al. (2010) reported that Swiss dairy farmers that did not follow *F. hepatica* control recommendations maintained a high percentage of positive fecal samples (32.5%) and seropositivity (62.1%) when compared to farms that had accepted the program protocol (9.3 and 21.4%, respectively). The authors pointed out that herds with a high prevalence of *F. hepatica* on the first visit also had a high prevalence on the second visit. In addition, the number of positive cattle detected on the first visit was associated with the

Table 3
Confusion matrix between the estimated classification and the samples during 2003–2010 for the incidence of *Fasciola hepatica* in Rio Grande do Sul, Brazil.

2003—overall accuracy = 61%				2004—overall accuracy = 55%			
Class—samples	Classification			Class—samples	Classification		
	Low	Medium	High		Low	Medium	High
Low (*n* = 109)	75 (69%)	28 (26%)	6 (5%)	Low (*n* = 76)	67 (88%)	8 (11%)	1 (1%)
Medium (*n* = 61)	15 (25%)	38 (62%)	8 (13%)	Medium (*n* = 99)	41 (41%)	52 (53%)	6 (6%)
High (*n* = 66)	7 (11%)	29 (44%)	30 (45%)	High (*n* = 88)	15 (17%)	47 (53%)	26 (30%)
User accuracy	77%	40%	68%	User accuracy	54%	49%	79%

2005—overall accuracy = 48%				2006—overall accuracy = 66%			
Class—samples	Classification			Class—samples	Classification		
	Low	Medium	High		Low	Medium	High
Low (*n* = 78)	61 (78%)	16 (21%)	1 (1%)	Low (*n* = 61)	38 (62%)	21 (35%)	2 (3%)
Medium (*n* = 104)	56 (54%)	46 (44%)	2 (2%)	Medium (*n* = 121)	28 (23%)	78 (65%)	15 (12%)
High (*n* = 107)	16 (15%)	64 (60%)	27 (25%)	High (*n* = 110)	15 (14%)	18 (16%)	77 (70%)
User accuracy	46%	39%	90%	User accuracy	47%	67%	82%

2007—overall accuracy = 52%				2008—overall accuracy = 63%			
	Classification				Classification		
Class—samples	Low	Medium	High	Class—samples	Low	Medium	High
Low (*n* = 84)	33 (39%)	48 (57%)	3 (4%)	Low (*n* = 110)	87 (79%)	23 (21%)	0 (0%)
Medium (*n* = 87)	29 (33%)	50 (58%)	8 (9%)	Medium (*n* = 89)	30 (34%)	56 (63%)	3 (3%)
High (*n* = 91)	8 (9%)	31 (34%)	52 (57%)	High (*n* = 55)	5 (9%)	34 (62%)	16 (29%)
User accuracy	47%	39%	83%	User accuracy	71%	50%	84%

2009—overall accuracy = 60%				2010—overall accuracy = 58%			
	Classification				Classification		
Class—samples	Low	Medium	High	Class—samples	Low	Medium	High
Low (*n* = 104)	56 (54%)	46 (44%)	2 (2%)	Low (*n* = 126)	82 (65%)	43 (34%)	1 (1%)
Medium (*n* = 100)	27 (27%)	65 (65%)	8 (8%)	Medium (*n* = 73)	22 (30%)	50 (69%)	1 (1%)
High (*n* = 35)	2 (6%)	11 (31%)	22 (63%)	High (*n* = 54)	9 (17%)	31 (57%)	14 (26%)
User accuracy	66%	53%	69%	User accuracy	73%	40%	88%

Table 4
Classification of non-pathogenic samples of *Fasciola hepatica* by the general model during 2003–2010 in Rio Grande do Sul, Brazil.

Year (n)	Low (%)	Medium (%)	High (%)
2003 (30)	23 (77%)	4 (13%)	3 (10%)
2004 (36)	30 (83%)	5 (14%)	1 (3%)
2005 (29)	27 (93%)	2 (7%)	0 (0%)
2006 (20)	15 (75%)	5 (25%)	0 (0%)
2007 (26)	10 (39%)	12 (46%)	4 (15%)
2008 (35)	27 (77%)	7 (20%)	1 (3%)
2009 (34)	19 (56%)	9 (26%)	6 (18%)
2010 (37)	27 (73%)	8 (22%)	2 (5%)
Average	22	6.5	2

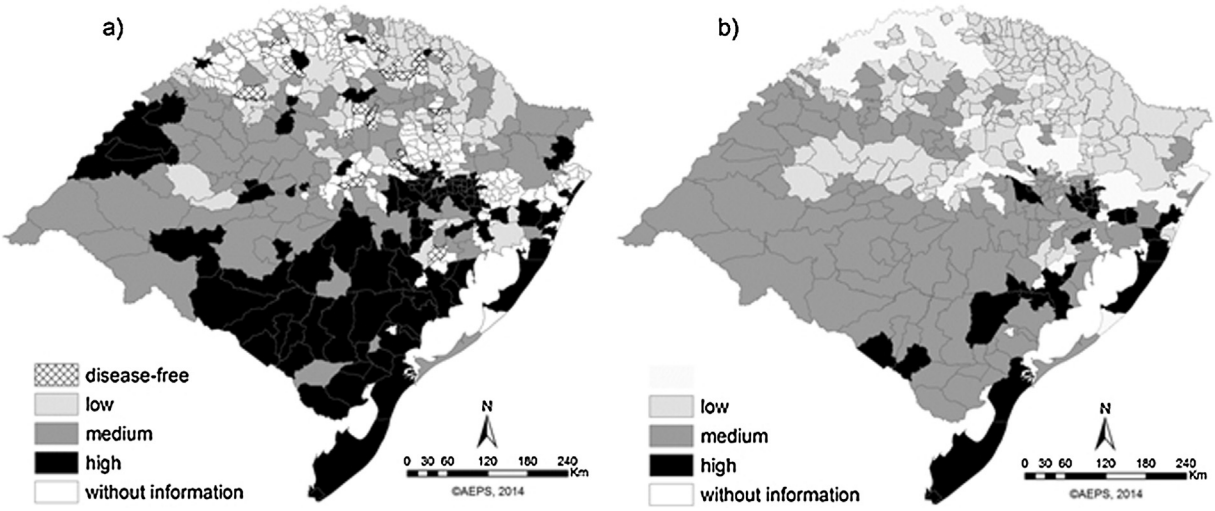


Fig. 2. Thematic maps. (a) Represents the observed Pi and (b) represents the estimated Pi by the general model for the occurrence of *Fasciola hepatica* in 2005.

level of seropositivity on the second visit ($p = 0.004$). As from our study, among the municipalities studied that had animals inspected at slaughtering for at least two consecutive years, 90% presented the disease in the following year and 80% presented the disease in the following two years after a positive result for fasciolosis. Bennema et al. (2011) also reported a stable prevalence and a cluster distribution of *F. hepatica* with small variations in between years in Flandres, Belgium.

Dutra et al. (2010) observed an inverse correlation (−0.80) between *F. hepatica* infection rate in cattle and the altitude in cities from the states of Paraná, Santa Catarina and Rio Grande do Sul. Our data indicates the importance of the Alt_ave variable, as it was the second highest Beta coefficient in the general model. Even though temperature can be a critical factor for the life cycle of *F. hepatica* completion, the T_{max}_win_ey variable ranked last in relevance in the present general model. Malone et al. (1998) determined that the greatest risk for *F. hepatica* occurrence was extended rainfall and soil moisture, as the risk of fascioliasis diminished in the driest and cooler areas of East Africa. Selemetas and Waal (2015) built a risk model in Ireland, where the most significant predictors were the total rainfall and total wet days. The risk model presented 100% sensitivity and 91% specificity and an accuracy of 95% correctly classifying cases.

The general model correctly classified between 48 and 66% of the samples with yearly variations. The low risk assessment resulted in the highest number of correctly classified samples as compared to the medium and high-risk assessment; it also presented few samples classified incorrectly as high risk. However, in some years the high-risk assessment presented few samples incorrectly classified and an expressive number of samples were incorrectly assessed as low risk. Martins-Bedê et al. (2009) also had similar results with *Schistosoma* sp. in the state of Minas Gerais, where the low-risk samples were correctly classified in greater numbers, when compared with the middle and high-risk samples with the two regression models. As such, the samples classified as high risk presented a great accuracy (68–90%). Martins et al. (2014) found similar evidences, correlating positive farms and previous cases of fascioliasis in the state of Espírito Santo, Brazil. We hypothesize that variations between the models can be attribute to a more constant weather conditions in Europe with less amplitude/variation than in South America.

The use of the proposed multiple linear regression model demonstrated the importance of including the year (time-based effect) for the construction of the model, considering two variables from the previous years (LnPi_ear and T_{max}_winear) as this disease has a survivorship that may last for years in the environment.

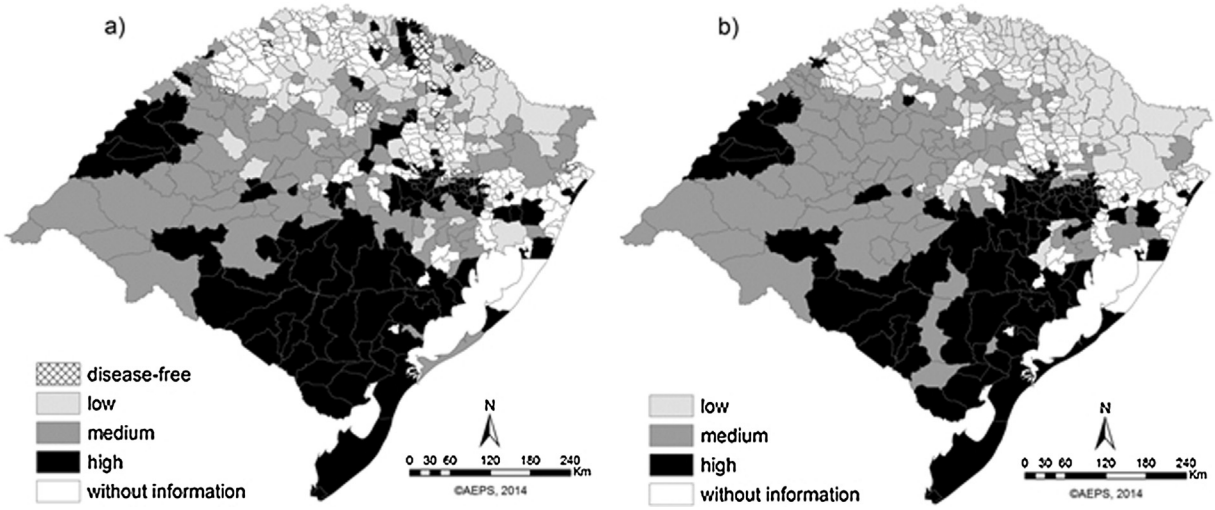


Fig. 3. Thematic maps. (a) Represents the observed Pi and (b) represents the estimated Pi by the general model for the occurrence of *Fasciola hepatica* in 2006.

The sequential values are important not only for epidemiological studies but to determine the disease dynamics, associated to climate and environmental variables that can vary along the year (Assis, 2011; Silva and Gurgel, 2011). The results evidenced a high accuracy percentage in low risk assessments, followed by the medium and high-risk samples. The overall results presented here were considered suitable taking into account the complexity of the disease and the limitation of more detailed available variables for the study. Although the environment and climate variables were used to explain the existing natural conditions for the occurrence of bovine fasciolosis, other important factors, such as farm management, soil type and treatment regime must also be considered in future studies to improve the disease estimation model.

Financial losses caused by bovine fascioliasis in the state of Rio Grande do Sul were measured to be of approximately US$ 147 million/year (Molento, MB, manuscript in preparation), thus, there is a need to identify the cities or regions facing a more critical scenario to implement vigilance programs based on the accuracy of our model. This general model can be used to predict scenarios for the following year based on values of the current year, showing a potential to be used as a tool for governmental and animal health policies. Thus, by following a control strategy designed according to a specific epidemiology model for a specific city in RS we expect that it will be possible to significantly reduce *F. hepatica* egg shedding and its prevalence in cattle. The proposed model would be useful in areas were fascioliasis is not an important issue and animals are still treated under a suppressive treatment strategy. In this case we want to significantly reduce the usage of anthelmintics and the selection pressure towards the development of drug resistance *F. hepatica*. We, the authors, will rely deeply in the good work of field practitioners at city and farm level to implement and follow appropriate control strategies.

There is also a growing concern about the lack of diagnosis of *F. hepatica* in humans in Brazil. Fascioliasis is acquired by the ingestion of food related to water and can produce fever, abdominal pain and intense bleeding in affected patients (Gupta, 2014). Although the parasite is one of the commonest found in the world causing an enormous public health problem, the WHO (2007) considered fascioliasis a neglected neglected disease due to its presence in developing countries (Peru, Bolivia, Vietnam, Egypt).

Although the incidence of tuberculosis in humans in Brazil has flattened out since 2005, the rate of the disease is close to 50 per 10,000 people, or 70 thousand new cases each year (WHO, 2013), 6.5 thousand only in the state of Rio Grande do Sul. The incidence of tuberculosis in cattle in Rio Grande do Sul is about 1.3% and there is a large geographical overlap for its occurrence in humans and animals, mostly in the South of the state and close to the state capital Porto Alegre (MS/SINAN/TB, 2015). Our concern also relates to the data from Claridge et al. (2012) that pointed out that the diagnostic of *F. hepatica* in cattle might have the possibility to have a 30% of false negative for the diagnostic for tuberculosis. Thus, as we are all aware, we must have a close look at disease dynamics (i.e., using multi regression analysis), and be prepared for all the challenges ahead of us.

This model can be applied in the Northern Argentina, Paraguay, Uruguay, Bolivia, as well as parts of Africa, Asia and Australia. All these sites meet the conditions for proliferation of the etiologic agent and the intermediate host for this disease. As such, the present model could also be used as a tool for large-scale decision makers in One World One Health related parasite control programs, assisting local communities (humans and animals) to better monitor any potential risk of infection.

Conflict of interest

All authors declare that there is no conflict of interest.

Acknowledgements

The authors wish to thank the Instituto Nacional de Pesquisas Espaciais, INPE personnel for their technical support and the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior, CAPES for funding this work and A.E.P. Silva, MSc scholarship. L.V.D. acknowledges CNPq Grant: 307666/2011-5.

References

Aleixo, M., Freitas, D.F., Dutra, L.H., Malone, J., Martins, I.V.F., Molento, M.B., 2015. *Fasciola hepatica*: epidemiology, perspectives in the diagnostic and the use of geoprocessing systems for prevalence studies. Semina 36, 1451–1466.

Assis, M.C., 2011. Abordagens espaciais para caracterização dos condicionantes socioambientais associados ao risco de malária em novas fronteiras na Amazônia: o caso de Lábrea, Amazonas. In: M.Sc. Dissertation in Remote Sensing. Instituto Nacional de Pesquisas Espaciais, São José dos Campos, INPE/451331-TDI, 138 pp.

Bennema, S.C., Ducheyne, E., Vercruysse, J., Claerebout, E., Hendrickx, G., Charlier, J., 2011. Relative importance of management, meteorological and environmental factors in the spatial distribution of *Fasciola hepatica* in dairy cattle in a temperate climate zone. Int. J. Parasitol. 41, 225–233.

Bennema, S.C., Scholte, R.G.C., Molento, M.B., Medeiros, C., Carvalho, O.S., 2014. *Fasciola hepatica* in bovines in Brazil: data availability and spatial distribution. Rev. Inst. Med. Trop. São Paulo 56, 35–41.

Claridge, J., Diggle, P., McCann, C.M., Mulcahy, G., Flynn, R., McNair, J., Strain, S., Welsh, M., Baylis, M., Williams, D.J.L., 2012. *Fasciola hepatica* is associated with the failure to detect bovine tuberculosis in dairy cattle. Nat. Commun. 3, 853, http://dx.doi.org/10.1038/ncomms1840.

Ducheyne, E., Charlier, J., Vercruysse, J., Rinaldi, L., Biggeri, A., Demeler, J., Brandt, C., Waal, T., Selemetas, N., Höglund, J., Kaba, J., Kowalczyk, S.J., Hendrickx, G., 2015. Modelling the spatial distribution of *Fasciola hepatica* in dairy cattle in Europe. Geospat. Health 9, 261–270.

Dutra, L.H., Molento, M.B., Naumann, C.R.C., Biondo, A.W., Fortes, F.S., Savio, D., Malone, J.B., 2010. Mapping risk of bovine fasciolosis in the south of Brazil using Geographic Information Systems. Vet. Parasitol. 169, 76–81.

Fonseca, F.R., 2009. Modelagem espacial de esquistossomose mansoni no estado de Minas Gerais, utilizando a conectividade de redes via estradas e rios. In: M.Sc. Dissertation in Remote Sensing. Instituto Nacional de Pesquisas Espaciais, São José dos Campos, INPE-16226-TDI/1545, 96 pp.

Freitas, D.F., Martins, I.V.F., Santos, G.M., Santos, A.R., Gomes, D.S., 2014. Bioclimatic distribution and prevalence maps for *Fasciola hepatica* in Espirito Santo State, Brazil. J. Venomous Anim. Toxins Incl. Trop. Dis. 20, 32.

Gupta, S.C., 2014. Fasciolosis in man and animals: An overview retrospect to historical perspective. National Symposium on Towards Food Security through Sustainable Animal Production and Integrated Parasite Management, 164–174.

Hasenack, H., Weber, E. (org.), 2010. Base cartográfica vetorial contínua do Rio Grande do Sul/escala 1:50.000. UFRGS, Ecol. Centre, Porto Alegre, DVD-ROM GIS series3.

Knubben-Schweizer, G., Rüegg, S., Torgerson, P.R., Rapsch, C., Grimm, F., Hässig, M., Deplazes, P., Braun, U., 2010. Control of bovine fasciolosis in dairy cattle in Switzerland with emphasis on pasture management. Vet. J. 186, 188–191.

Malone, J.B., Gommes, R., Hansen, J., Yilma, J.M., Slingenberg, J., Snijders, F., Nachtergaele, F., Ataman, E., 1998. A geographic information system on the potential distribution and abundance of *Fasciola hepatica* and *F. gigantica* in east Africa based on Food and Agriculture Organization databases. Vet. Parasitol. 78, 87–101.

Martins, I.V.F., Avelar, B.R., Bernardo, C.C., Leão, A.C., Pereira, M.J.S., 2014. Distribution of bovine fasciolosis and associated factors in south Espírito Santo, Brazil: an update. Braz. J. Vet. Parasitol. 23, 23–29.

Martins, F.T., 2008. Mapeamento do risco da esquistossomose no Estado de Minas Gerais, usando dados ambientais e sociais. In: MSc Dissertation. Applyed Computing. Instituto Nacional de Pesquisas Espaciais, São José dos Campos, INPE-16228-TDI/1547, 144 pp.

Martins-Bedê, F.T., Freitas, C.C., Dutra, L.V., Sandri, S.A., Fonseca, F.R., Drummond, I.N., Guimarães, R.J.P.S., Amaral, R.S., Carvalho, O.S., 2009. Risk mapping of schistosomiasis in Minas Gerais, Brazil, using MODIS and Socioeconomic Spatial Data. IEEE Trans. Geosci. Remote Sens. 47, 3899–3908.

Mas-Coma, M.S., Esteban, J.G., Bargues, M.D., 1999. Epidemiology of human fascioliasis: a review and proposed new classification. Bull. World Health Org. 77, 340–346.

Neter, J., Kutner, M.H., Nachtssheim, C.J., Wasserman, W. (Eds.), 1996. Applied Linear Statistical Models. WCB McGraw-Hill, Boston, p. 1408pp.

Oliveira, S.M., Filha, E.S., 2009. Inst. Biologico 71, 5–7.

Selemetas, N., Waal, T., 2015. Detection of major climatic and environmental predictors of livers fluke exposure risk in Ireland using spatial cluster analysis. Vet. Parasitol. 209, 242–253.

Silva, A.E.P., Gurgel, H.C., 2011. Leishmaniose tegumentar Americana e suas relações sócio ambientais no município de Ubatuba-SP. Confins (Paris) 13, 1–18.
Toledo, R., Esteban, J.G., Fried, B., 2012. Current status of food-borne trematode infections. Eur. J. Clin. Microbiol. Infect. Dis. 31, 1705–1718.
Valencia-López, N., Malone, J.B., Carmona, C.G., Velásquez, L.E., 2012. Climate-based risk models for *Fasciola hepatica* in Colombia. Geospat. Health 6, 67–85.
WHO, World Health Organization, 2007. Action against worms. Newsletter 10, 7.
WHO, World Health Organization, 2013 Global Tuberculosis Report.
WHO, World Health Organization, 2014. Neglected Tropical Diseases. A statistical update: latest data available. WHO/HTM/NTD. Geneva, Switzerland. 29 pp.